

The Market Pricing of Proprietary AI Model Capability

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Abstract

Which AI model capability measures do financial markets price? We study 60 major AI model releases from 2024 to early 2026 and a firm-event panel of 86 AI-related listed firms. Across a specification curve, the broad Artificial Analysis Intelligence Index has the strongest long-window signal; narrower coding and math indexes do not. In closed-source releases, a one-standard-deviation increase in this index is associated with a 3.0 percentage point higher CAR[0, +20] ($\hat{\beta} = 0.0023$, wild bootstrap $p = 0.008$). The slope is weaker for open-weight releases and survives firm fixed effects, Fama-French abnormal returns, event-level aggregation, and creator exclusions. Markets price broad AI capability when releases preserve proprietary rents.

JEL codes: G14, O30, L86

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1 Introduction

AI model releases now move a large part of the technology sector’s information environment. These releases differ from ordinary product announcements. A model release discloses a new capability frontier, and that frontier can matter for cloud demand, software margins, developer ecosystems, and competitive position. The financial question is which part of that capability frontier equity markets price.

The answer should depend on appropriability. A high-capability closed-source model can support API revenue, customer lock-in, and ecosystem rents. A high-capability open-weight model diffuses similar technical performance more broadly. The same benchmark score can therefore imply different cash-flow news, depending on whether the releasing ecosystem can capture the rents from the model.

This paper tests that idea using 60 major AI model release events from January 2024 through March 2026. We match releases to 86 AI-related listed firms and estimate firm-event abnormal returns. The main outcome is the market-model cumulative abnormal return over $[0, +20]$ trading days. We compare four Artificial Analysis (AA) measures: Intelligence Index, Coding Index, Math Index, and Media Elo. Because capability varies at the event level, we cluster inference by event and report CR2 small-cluster corrections and Rademacher wild cluster bootstrap p -values.

The metric comparison is itself part of the result. The broad AA Intelligence Index has the strongest and most economically meaningful long-window relation with returns. In $CAR[0, +20]$ specifications, 47.2 percent of AA Intelligence estimates are significant at 5 percent and 83.3 percent are positive. Coding and math indexes are much weaker. Media Elo has some significant specifications, but its median long-window coefficient is slightly negative and its strongest evidence appears in shorter windows. Thus the evidence does not say that every AI benchmark is priced. It says that the broad benchmark that combines text, reasoning, code, and multimodal performance is the relevant long-window pricing signal.

The main result is concentrated in proprietary releases. For closed-source releases, a

one-point increase in the AA Intelligence Index predicts a 0.23 percentage point higher CAR[0, +20] ($\hat{\beta} = 0.0023$, CR2 $p = 0.007$, wild $p = 0.008$). The index has a standard deviation of 13.1 points, so the estimate implies a 3.0 percentage point higher abnormal return for a one-standard-deviation capability increase. The full-sample slope is positive but only borderline under small-cluster inference. This pattern points to a selective capability-pricing channel rather than a general market reward for model performance.

Open-weight releases attenuate the capability slope. In the interaction model, the closed-source slope is 0.0023, while the interaction between intelligence and open-weight status is -0.0037. The interaction is statistically significant under CR2 ($p = 0.048$) and weaker under the wild bootstrap ($p = 0.086$). We interpret this result as mechanism evidence, not as a stand-alone open-source effect. At average capability, open-weight releases do not have a statistically different mean return from closed-source releases; the difference lies in the marginal return to additional capability.

Several checks support the closed-source result. Adding firm fixed effects leaves the closed-source coefficient at 0.0023 and significant under event, firm, and two-way clustering. Replacing market-model returns with Fama-French three-factor abnormal returns keeps the coefficient positive and significant. Aggregating to the event level removes the repeated-firm panel structure; the closed-source event-level coefficient remains positive ($\hat{\beta} = 0.0017$, $p = 0.016$). The result is also robust to excluding Google, OpenAI, Anthropic, Alibaba, Meta, xAI, or Microsoft one at a time. We do not find a statistically significant relation between intelligence and pre-event CARs, which reduces concern that the main result only reflects a pre-announcement run-up.

The paper relates to three literatures. Event studies show how public announcements reveal value-relevant information (MacKinlay, 1997). Work on innovation and market value shows that technological advances are priced when firms can capture rents from them (Teece, 1986; Hall, Jaffe, and Trajtenberg, 2005). Research on media and financial markets shows that attention and sentiment can move prices (Tetlock, 2007; Engelberg and Parsons, 2011).

We bring these ideas to AI model releases by measuring technical capability directly and by showing that the return relation is strongest for broad capability and when the release remains proprietary.

2 Data and Empirical Design

The unit of observation is a firm-event pair. We begin with 60 major AI model release events between January 2024 and March 2026. Events enter the sample when they are official model or model-family releases, have a confirmed public release date, are accessible or publicly benchmarked, and can be linked to at least one AI-related listed firm. The clean panel has 5,160 firm-event observations. The main regression sample has 3,780 observations, 47 releases with nonmissing AA Intelligence Index scores, and 85 listed firms with complete controls. The closed-source sample has 2,899 observations across 36 events. The open-weight sample has 881 observations across 11 events.

We estimate abnormal returns using a market model fitted over a pre-event estimation window. The main outcome, $CAR[0, +20]$, sums abnormal returns from the release date through 20 trading days after the release. AI releases are clustered in calendar time, so this window should be read as medium-window repricing in a dense news environment. The online appendix shows that the results are similar with Fama-French three-factor abnormal returns and reports pre-event CAR tests.

The main capability proxy is the AA Intelligence Index, a benchmark-based measure for text, reasoning, code, and multimodal models. We use it because it is the broadest technical capability measure in the data, not because it is the only available benchmark. The online appendix reports the same specification-curve exercise for the AA Coding Index, AA Math Index, and AA Media Elo. We center the AA Intelligence Index at its sample mean, 26.6, when estimating interaction models. Firm controls are log assets, book-to-market, prior-quarter volatility, and momentum. All specifications include year fixed effects. Some

robustness checks add firm fixed effects.

We estimate variants of

$$\text{CAR}_{ie} = \beta \text{Capability}_e + \mathbf{X}'_{ie} \gamma + \alpha_t + \varepsilon_{ie}, \quad (1)$$

where i indexes firms and e indexes release events. We estimate this equation for each capability measure, then focus on the AA Intelligence Index where the long-window signal is strongest. We also interact intelligence with an open-weight indicator. Standard errors are clustered by event. The appendix reports CR2 and wild cluster bootstrap inference because the effective variation comes from 47 release events.

3 Results

Table 1 compares the candidate capability measures. AA Intelligence has the largest share of significant estimates, the largest share of positive estimates, and the largest median coefficient. The narrower coding and math indexes do not show the same pattern. AA Media Elo is positive in some specifications, but its median $\text{CAR}[0, +20]$ coefficient is slightly negative. The rest of the paper therefore treats AA Intelligence as the primary broad- capability measure and treats the other metrics as comparison benchmarks.

Table 1: Which capability metric carries the pricing signal?

Capability measure	Specifications	5% significant	Positive	Median coefficient
AA Intelligence Index	36	47.2%	83.3%	0.000848
AA Media Elo	33	27.3%	45.5%	-0.000085
AA Coding Index	36	25.0%	69.4%	0.000334
AA Math Index	33	9.1%	51.5%	0.000004

The table summarizes $\text{CAR}[0, +20]$ specifications that vary the control set, sample split, and year fixed effects. The reported shares use the clustered standard error from each specification. The median coefficient is over all $\text{CAR}[0, +20]$ specifications for the stated metric.

Table 2 shows the core AA Intelligence estimates. Column 1 pools all capability events and gives a positive but marginal slope under conservative inference. Column 2 restricts the

sample to closed-source releases. The coefficient rises to 0.0023 and remains significant under CR2 and wild bootstrap inference. Column 3 shows the interaction model. The closed-source slope remains positive; the open-weight interaction is negative.

Table 2: AI model capability and post-release abnormal returns

	Full sample (1)	Closed-source (2)	Interaction (3)
AA Intelligence Index	0.00152 (0.00067)	0.00232 (0.00062)	0.00226 (0.00060)
CR2 p -value	0.054	0.007	0.005
Wild p -value	0.053	0.008	0.008
AA Intelligence $\times$ Open-weight			-0.00373 (0.00120)
CR2 p -value			0.048
Wild p -value			0.086
Year fixed effects	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes
Observations	3,780	2,899	3,780
Events	47	36	47

The dependent variable is market-model CAR[0, +20]. Column 3 uses the AA Intelligence Index centered at its sample mean. Standard errors in parentheses are clustered by release event. CR2 reports Bell-McCaffrey small- cluster p -values. Wild reports Rademacher wild cluster bootstrap p -values with 4,999 draws.

Table 3 reports the most important robustness checks. Firm fixed effects leave the closed-source coefficient unchanged. Two-way clustering by event and firm also leaves it significant. Using Fama-French abnormal returns reduces the coefficient but keeps it positive. The pre-event CAR coefficient is not statistically different from zero. At the event level, the closed-source mean-CAR coefficient remains positive and significant. The result also survives a rough restriction to US-traded tickers.

Media sentiment does not explain away the capability result. Table 4 shows that FinBERT sentiment is negatively associated with post-release returns. In the joint model, both sentiment and capability retain their signs, although both are marginal under CR2. One interpretation is that strongly positive media coverage captures already anticipated releases that partly reverse after the announcement. We treat this as a competing-information check rather than

Table 3: Robustness of the closed-source capability slope

Specification	Coefficient	SE	p -value	Events
Firm fixed effects, event cluster	0.00226	0.00063	0.001	36
Firm fixed effects, firm cluster	0.00226	0.00070	0.002	36
Firm fixed effects, two-way cluster	0.00226	0.00087	0.014	36
Fama-French CAR[0, +20]	0.00118	0.00051	0.027	36
Pre-event CAR	0.00047	0.00030	0.119	36
Event-level mean CAR	0.00172	0.00067	0.016	36
US-traded ticker screen	0.00271	0.00072	0.001	36

The table reports the coefficient on the AA Intelligence Index for closed-source releases. The pre-event CAR row uses the pre-release abnormal return as the dependent variable. The event-level row aggregates CAR[0, +20] to one equal-weighted mean return per release. The US-traded ticker screen uses a rule-based ticker filter and is reported as a robustness check, not as a formal exchange-country classification.

as a fully identified sentiment mechanism.

Table 4: Capability and media sentiment

	Sentiment only (1)	Sentiment only (2)	Joint (3)
FinBERT sentiment, ± 5 days	-0.081		-0.061
CR2 p -value	0.003		0.057
FinBERT sentiment, ± 20 days		-0.096	
CR2 p -value		0.015	
AA Intelligence Index			0.00158
CR2 p -value			0.087
Year fixed effects	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes
Observations	3,780	3,780	3,780
Events	47	47	47

The dependent variable is market-model CAR[0, +20]. Sentiment is the event-level mean FinBERT score. The joint model includes both sentiment and the AA Intelligence Index. Event-level sentiment and intelligence are not statistically related in the sample ($p = 0.317$).

4 Conclusion

AI model capability is not priced uniformly across metrics or release regimes. The strongest long-window signal comes from the broad AA Intelligence Index, not from narrower coding or

math measures. Within that broad metric, the strongest evidence appears for closed-source releases, where a one-standard-deviation increase predicts a 3.0 percentage point higher 20-day abnormal return. Open-weight releases weaken that slope. The evidence is consistent with a simple finance mechanism. Broad technical capability becomes value-relevant when firms can capture rents from it.

The result survives firm fixed effects, alternative abnormal-return models, event-level aggregation, and creator exclusions. It does not appear in pre-event CARs. These checks do not make model releases random shocks. They do show that the closed-source capability slope is not an artifact of firm composition, a single developer, or the market-model return benchmark. In this setting, markets appear to distinguish proprietary technical progress from AI news more broadly.

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